

Quiz-2: PCA & MBA

1. PCA extracts the variance structure from high dimensional data such that the variance of projected data is minimized. (T/F)
2. The principal components obtained from PCA are always independent of each other. (T/F)
3. Ignoring the components of small eigenvalues will not lose information. (T/F)
4. When we use PCA, we need data to be labelled (T/F)
5. PCA is useful in removing noise and redundant features from high-dimensional datasets. (T/F)
6. What is the main goal of PCA?
 - A. Reduce the number of dependent variables
 - B. Minimize the variance of the dataset
 - C. Maximize the variance captured by fewer components
 - D. Increase the number of dimensions
7. The first principal component (PC1) is:

- A. The direction of minimum variance
- B. The direction of maximum variance
- C. Perpendicular to the second principal component
- D. A combination of dependent variables

8. What is the "Curse of Dimensionality" in PCA?

- A. As dimensions increase, data points become closer to each other
- B. As dimensions increase, data points become more sparse and distant
- C. PCA becomes more effective as dimensions increase
- D. PCA cannot be used for dimensionality reduction

9. What happens to correlated features in PCA?

- A. They remain unchanged
- B. They are projected onto different dimensions
- C. They are merged into principal components
- D. They are removed from the dataset

10. What is the effect of normalizing data before applying PCA?

- A. Increases the impact of high-variance variables
- B. Ensures that all variables contribute equally
- C. Reduces the number of principal components
- D. Increases computation time

11. Which of the following is/are true about PCA?

- i. PCA is an unsupervised method of optimization
- ii. It searches for the directions that data have the largest variance
- iii. Maximum number of principal components $\leq$ number of features
- iv. All principal components are orthogonal to each other

- A. All to the above
- B. i) and ii)
- C. i), ii) and iii)
- D. i), ii) and iv)

12. Which statement is incorrect about PCA?

- A. PCA requires standardization because we must assign equal weights to all variables; otherwise, we may receive misleading recommendations.
- B. Rotation is a crucial PCA phase that entails changing the coordinate system for the primary components. If we do not rotate the components, the effect of PCA will be diminished.
- C. PCA is a statistical technique for reducing the dimensionality of a dataset. This is accomplished by linearly transforming the data into a new coordinate system where (most of) the variation in the data can be described with fewer dimensions than the initial data.
- D. PCA reduces the dimensionality of a dataset by non-linearly transforming the data into a new coordinate system where (most of) the variation in the data can be described with fewer dimensions than the initial data.

13. If all principal components have the same eigenvalue, what does this indicate?

- A. Strong correlation among variables
- B. No significant variance in the data
- C. Equal contribution of all components
- D. Outliers in the dataset

14. Which of the following is False about principal components analysis (PCA)? Assume that no two eigenvectors of the sample covariance matrix have the same eigenvalue.

- A. Appending a 1 to the end of every sample point doesn't change the results of performing PCA (except that the useful principal component vectors have an extra 0 at the end, and there's one extra useless component with eigenvalue zero).
- B. If you use PCA to project d -dimensional points down to j principal coordinates, and then you run PCA again to project those j -dimensional coordinates down to k principal coordinates, with $d > j > k$, you always get the same result as if you had just used PCA to project the d -dimensional points directly down to k principal coordinates.
- C. If you perform an arbitrary rigid rotation of the sample points as a group in feature space before performing PCA, the principal component directions do not change.
- D. If you perform an arbitrary rigid rotation of the sample points as a group in feature space before performing PCA, the largest eigenvalue of the sample covariance matrix does not change

15. In Scikit-learn's PCA implementation, what does the parameter `n_components=2` specify?

- A. The number of original features to keep
- B. The number of principal components to retain
- C. The maximum variance explained
- D. The number of rows in the dataset

16. After applying PCA with `n_components=2`, what will be the shape of `X_transformed` if the original dataset `X` had 500 samples and 10 features?

- A. (500, 10)
- B. (10, 2)
- C. (500, 2)
- D. (2, 500)

17. A data point $(x, y) = (3, 4)$ is projected onto the principal component $v = (0.6, 0.8)$. What is the projected coordinate along this PC?

- A. 5
- B. 4.8
- C. 6.5
- D. 3.2

18. A dataset has 3 variables with total variance of 15. After applying PCA, the first two principal components explain 90% of the variance. How much variance is captured by the third principal component?

- A. 0.5
- B. 1.5
- C. 3.0
- D. 1.0

19. Suppose we perform PCA on a dataset and obtain two principal components with eigenvalues 5 and 2. What percentage of the total variance is explained by the first principal component?

- A. 30%

- B. 50%
- C. 71.4%
- D. 80%

20. Market Basket Analysis is widely used in:

- A. Medical image processing
- B. Social media sentiment analysis
- C. Retail and e-commerce analytics
- D. Cryptography

21. If an association rule has a high lift value, it means:

- A. The rule is weak
- B. The products are independent
- C. The presence of one product significantly increases the likelihood of purchasing another
- D. The rule is not valid

22. What is the main goal of Market Basket Analysis?

- A. Reduce the number of products in a store
- B. Discover relationships between items frequently purchased together
- C. Identify outliers in sales data
- D. Perform time series forecasting

23. Which measure represents the likelihood that a product is purchased given another product?

- A. Support
- B. Confidence
- C. Lift

D. Correlation

24. In Market Basket Analysis, what does support measure?

- A. The proportion of transactions containing a specific item or itemset
- B. The strength of the association rule
- C. The percentage of unique customers buying a product
- D. The time taken for a customer to make a purchase

25. A high confidence value in an association rule means:

- A. The rule is strong
- B. The rule has no significance
- C. The rule is based on probability
- D. The rule must always hold

26. If the lift value of a rule is equal to 1, it means:

- A. The items are strongly correlated
- B. The items are independent
- C. The items always appear together
- D. The rule is invalid

27. A supermarket dataset contains 1,000 transactions. 200 transactions include "Milk". What is the support for the item "Milk"?

- A. 0.02
- B. 0.2
- C. 0.5
- D. 0.8

28. A Market Basket Analysis finds that 150 transactions include "Bread" and 100 include both "Bread and Butter." What is the confidence of the rule: Bread → Butter?

- A. 0.33

- B. 0.5
- C. 0.67
- D. 0.75

29. A store finds that:

- $\text{Support}(\text{Milk}) = 0.4$
- $\text{Support}(\text{Bread}) = 0.5$
- $\text{Support}(\text{Milk \& Bread}) = 0.3$

What is the **Lift** of the rule **Milk** $\rightarrow$ **Bread**?

- A. 1.2
- B. 1.5
- C. 0.75
- D. 2.0

30. A dataset contains 5,000 transactions. If a store wants to set a minimum support threshold of 5%, how many transactions must contain an item to be considered frequent?

- A. 100
- B. 250
- C. 500
- D. 1,000